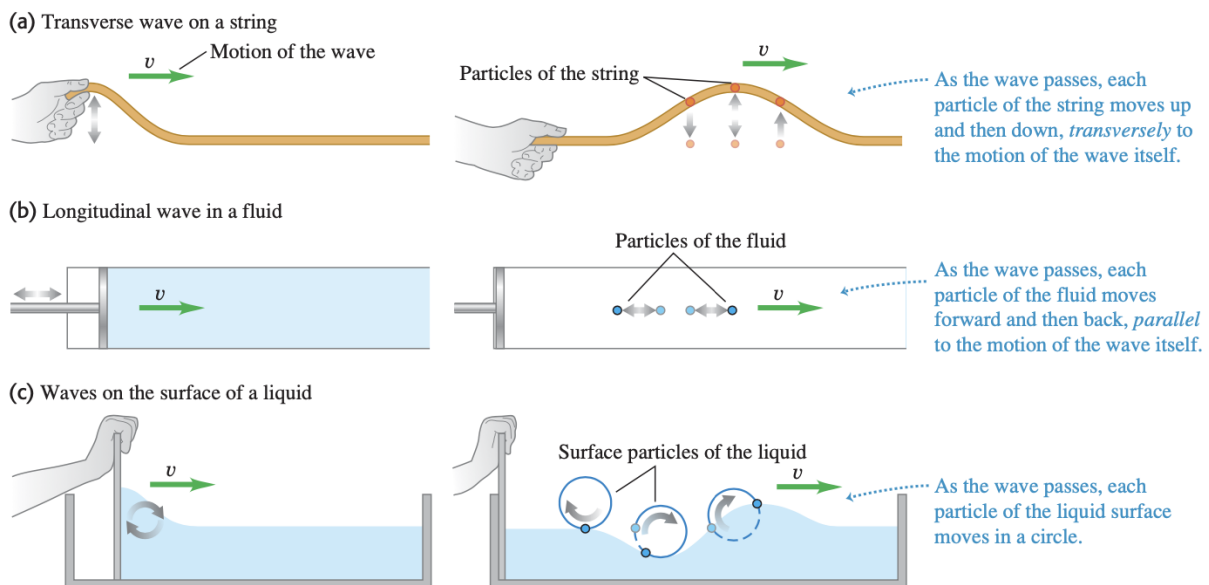


PHY1001: Mechanics (Week 12-14)

In this chapter, we will discuss waves, which is defined as the **oscillation (in time) that travels in space and transfers energy from one place to another**. The oscillation with time and wave propagation are described by the wave equation.

When the displacement of the medium are perpendicular or transverse to the propagation direction of the wave along the medium, we define this type of wave as the transverse wave. For example, the wave travels along a string and the electromagnetic wave (light wave) are transverse waves.¹

When the displacement of the medium are along the same direction as the propagation direction of the wave, we name this type as the longitudinal wave. For example, the sound wave is a longitudinal wave. Earthquake waves have both longitudinal and transverse components.



Three ways to make a wave that moves to the right: (a) The hand moves the string up and then returns, producing a transverse wave. (b) The piston moves to the right, compressing the gas or liquid, and then returns, producing a longitudinal wave. (c) The board moves to the right and then returns, producing a combination of **longitudinal and transverse** waves.

1 Waves in general

Usually, there are three types of waves.

1. A mechanical wave is a **disturbance that travels through some material or substance** called the medium for the wave. **Ripples on a pond, musical sounds, earthquakes** (seismic tremors) are all wave phenomena. As the wave travels through the medium, particles which make up the medium undergo displacement and carry energy as the wave propagates along the direction of the wave.
2. Electromagnetic waves—including light, radio waves, infrared and ultraviolet radiation, and X rays—can propagate even in empty space (vacuum), where there is no medium. These waves satisfy the electromagnetic wave equation derives from Maxwell's equations and travel at speed of light in vacuum.
3. Matter waves. In quantum mechanics, according to the de Broglie hypothesis (1924), there are waves

are associated with electrons, protons, and other fundamental particles, and even atoms and molecules. All matter exhibits properties of both particles and waves, and there is a relationship between the wavelength, λ and the momentum, p , through the Planck constant, $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$:

$$\lambda = \frac{h}{p}. \quad (1)$$

Because we commonly think of these particles as constituting matter, such waves are called matter waves. The wave equation for the matter wave in QM is known as the the Schrödinger equation.

Wave-particle duality: every particle may be partly described in terms of not only particles, but also of waves. Neither particle nor wave can fully explain the physics, but together they do. This is the **quantum nature of physical entities**. They are like two **sides of the same coin**, or two faces of the same reality. We have to look at the particle (object) from different angles to get the full picture of the reality, because we are living in the same reality.

¹Strictly speaking electromagnetic waves include light, radio waves, X rays, gamma rays, and microwaves, among others. The various types of electromagnetic waves differ only in wavelength and frequency.

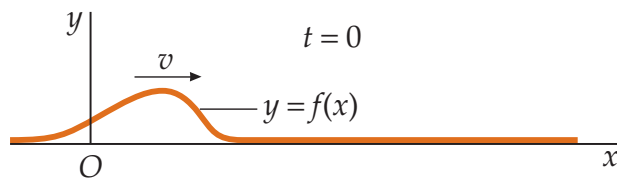


In this course, we will only focus on the **Mechanical waves** which have three interesting properties:

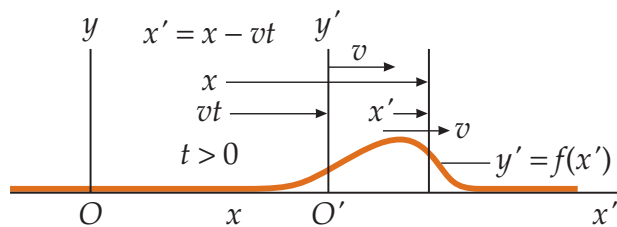
- The disturbance travels or propagates with a definite speed through the medium, which is called the speed of propagation, or simply the wave speed v .
- The medium itself does not travel through space; its individual points undergo transverse or longitudinal motions around their equilibrium positions. It is the overall pattern or shape of the wave disturbance is what travels.
- Last but not least, to create a disturbance, we have to put in energy by doing work on the system, and let the wave motion transport this energy as the wave travels. It is important to note that waves transport energy, but not matter, from one region of space to another.

Wave properties

Wave function: **The displacement of the medium from the EP as the function of space and time (x and t).**



(a)



(b)

Above figure (a) shows a pulse (disturbance) on a string at $t = 0$. The shape of the string at this moment can be described by the function $y = f(x)$. At some later time, the pulse is further down the string which can be represented by the function $y = f(x - vt)$. The most convenient way to view the propagation of the wave pulse is to change coordinate system to O' in which $x' = x - vt$. The string is described in this frame by $f(x')$ for all times. Thus for **fixed point on the pulse $x' = x - vt = \text{constant}$** , it travels at speed of v in the O frame as shown below

$$\frac{dx'}{dt} = 0 = \frac{dx}{dt} - v, \Rightarrow \frac{dx}{dt} = v. \quad (2)$$

Another way is to look at the peak of the pulse. Suppose the peak of $f(x)$ is at $x = x_m$, then the peak position moves to $x_m + vt$ after time t . Therefore, the function

$y = f(x - vt)$ describes a pulse moving to the right. Similarly, $y = f(x + vt)$ describes a pulse moving to the left. They both satisfies the differential wave equation

$$\frac{1}{v^2} \frac{\partial^2 y}{\partial t^2} = \frac{\partial^2 y}{\partial x^2}. \quad (3)$$

A few comments are in order.

1. Traveling wave solutions

$y = f(x - vt)$ moving in the $+x$ direction, (4)

$y = f(x + vt)$ moving in the $-x$ direction. (5)

2. v is the speed of propagation of the wave. (Because v is a speed and not a velocity, it is always a positive quantity.) The function $y = f(x - vt)$ is called a wave function.

3. For waves on a string, the wave function represents the transverse displacement of the string.

4. For sound waves in air, the wave function can be the longitudinal displacement of the air molecules, or the pressure of the air.

Speed of waves

Interestingly, it turns out that for many different type of mechanical waves, the expression for wave speed has the general form

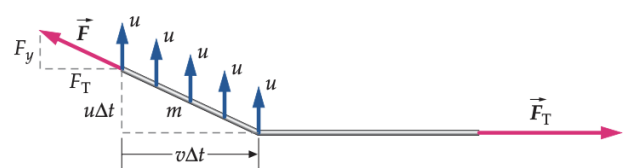
$$v = \sqrt{\frac{\text{Restoring Force}}{\text{Inertia}}} \quad (6)$$

1.0.1 Transverse wave

A general property of waves is that their speed relative to the medium depends on the properties of the medium, but is independent of the motion of the disturbance. For wave pulses on a taut rope, **the speed of wave is only depending on the tension of the rope and linear mass density of the rope.** If F_T is the tension (we use F_T rather than T for tension because we use F_T for the period) and μ is the linear mass density (mass per unit length), then the wave speed is

$$v = \sqrt{\frac{F_T}{\mu}}. \quad (7)$$

F_T plays the role of the restoring force, since it tends to bring the string back to its undisturbed, equilibrium configuration. The linear mass density μ provides the inertia.



There are at least three ways to derive the above wave speed.



1. Consider the propagation of the wave front movement as shown above

$$\underbrace{F_y \Delta t = F_T \Delta t \frac{u}{v} = \Delta m u = (\mu v \Delta t) u}_{\text{Propagation of the wave front}} \Rightarrow v = \sqrt{\frac{F_T}{\mu}}. \quad (8)$$

As the end of the string moves upward at constant speed u , the point where the string changes from horizontal to inclined moves to the right at the wave speed v .

2. Consider the centripetal acceleration of the wave crest (or trough) in the co-moving frame with speed v as illustrated in the textbook. $2F_T \frac{\Delta l}{2R} = (\mu \Delta l) \frac{v^2}{R}$.
3. Consider an arbitrary segment of the string as discussed below.

1.0.2 Longitudinal wave

Sound propagates due to the compression of the medium along the direction of the propagation. Similarly, the speed of a longitudinal sound wave in a fluid (including gas and liquid) is given by

$$v_s = \sqrt{\frac{B}{\rho}}, \quad (9)$$

where $B \equiv -V \frac{dP}{dV} = \frac{\text{—Pressure Change}}{\text{Fractional Volume Change}}$ is the bulk modulus. B is the measure of how resistant to compression that substance is. The bulk modulus tells us exactly how large the "restoring force" is against compression.

In addition, the expression for the bulk modulus of the ideal gas is $B = \gamma P$.

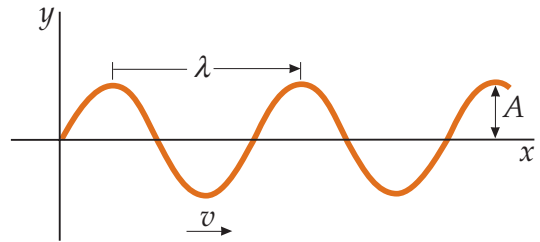
$$v_s = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma R T}{M}}, \quad (10)$$

where P is the pressure of the gas, $R = 8.31 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$ is the molar gas constant and T is the temperature. M is the average molar mass air is $29 \times 10^{-3} \text{ kg/mol}$. Roughly speaking, the gas pressure provides the force while the inertia is given by the density of the gas. This ratio $\gamma = 5/3$ for an ideal monoatomic gas and $\gamma = 7/5$ for air, which is predominantly a diatomic gas. (The major constituents of Earth atmosphere are nitrogen and oxygen.) Sir Newton missed the γ factor which results in a speed off by about 15%, when he first studied this problem. Laplace rectified this error by correctly noticing that sound wave compression and expansion of air is an adiabatic process, not an isothermal process.

It is useful to use the above equation to estimate the speed of sound in air and find $v_s \approx 350 \text{ m/s}$ in air. Sound waves tend to travel faster in water. It is around 1500 m/s , which is 4.3 times as fast as in air.

2 Harmonic (sinusoidal) waves

A more interesting situation develops when we give the free end of the string a repetitive, or periodic, motion. Then each particle in the string also undergoes periodic motion as the wave propagates, and we have a **periodic wave**. As shown below, if each point of the medium moves with simple harmonic motion, we call the corresponding wave the **harmonic wave**. Harmonic wave is one of the most common periodic waves.



The minimum distance after which the wave repeats itself is called the **wavelength λ** . Similarly, the periodicity in time is called the **period T** . So the speed is also given by

$$v = \frac{\lambda}{T} = f\lambda. \quad (11)$$

Since the above relations arise only from the definition of wavelength and frequency, it applies to all kinds of periodic waves. The **displacement $y(x, t)$** of a harmonic wave traveling along $+x$ direction is described by

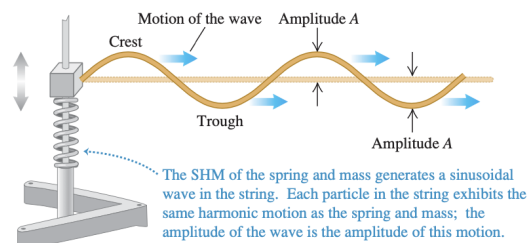
$$y(x, t) = A \sin(kx - \omega t) = A \sin\left[k\left(x - \frac{\omega}{k}t\right)\right] \quad (12)$$

where $k = \frac{2\pi}{\lambda}$ is called the (angular) **wave number** which has the dimension of L^{-1} and $\frac{\omega}{k} = \frac{2\pi}{T} \frac{\lambda}{2\pi} = v$ with the **angular frequency ω** . This wave is propagating along the $+x$ direction, since the position $x = vt + x_0$ in the wave with the same y displacement $A \sin kx_0$ increases with t . In contrast, $y(x, t) = A \sin(kx + \omega t)$ describes the wave traveling in the $-x$ direction.

Harmonic waves are periodic in both space (x) and time (t).

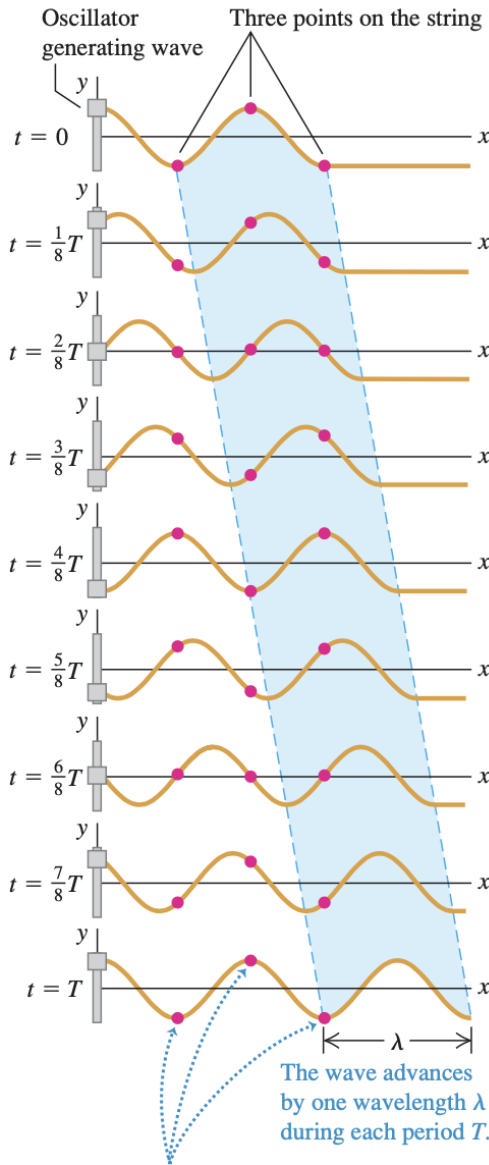
- **Periodic in space x :**
 $y(x + \lambda, t) = A \sin(kx + 2\pi - \omega t) = y(x, t)$.
- **Periodic in time t :**
 $y(x, t + T) = A \sin(kx - \omega t - 2\pi) = y(x, t)$.

The sinusoidal wave generated by SHM advances steadily toward the right, as indicated by the highlighted area.





The string is shown at time intervals of $\frac{1}{8}$ period for a total of one period T . The highlighting shows the motion of one wavelength of the wave.



Each point moves up and down in place. Particles one wavelength apart move in phase with each other.

The transverse velocity and acceleration are then given by

$$v_y(x, t) = \frac{\partial y(x, t)}{\partial t} = -\omega A \cos(kx - \omega t), \quad (13)$$

$$a_y(x, t) = \frac{\partial^2 y(x, t)}{\partial t^2} = -\omega^2 A \sin(kx - \omega t) \quad (14)$$

$$= -\omega^2 y(x, t), \quad (15)$$

respectively. Here ∂t is called **partial derivative w.r.t. time t** (when t varies), while the other variable x is constant because we are looking at a particular point on the string. The **partial derivative** is a simple extension of the derivative.

We can also compute partial derivatives of $y(x, t)$ with respect to x , holding t constant. The first derivative $\partial y(x, t)/\partial x$ is the slope of the string at point x and at time t . The second partial derivative with respect to x is the

curvature of the string:

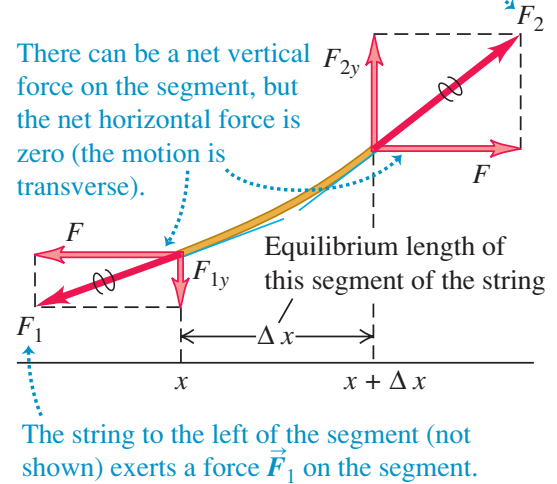
$$\frac{\partial^2 y(x, t)}{\partial x^2} = -k^2 A \sin(kx - \omega t) \quad (16)$$

$$= -k^2 y(x, t) = \frac{k^2}{\omega^2} a_y = \frac{a_y}{v^2}, \quad (17)$$

which means that a_y in a string is tightly related to the **curvature of the wave**!

Advanced topic: the wave equation

The string to the right of the segment (not shown) exerts a force $\vec{F}_2$ on the segment.



Consider the free-body diagram for a segment Δx of string, we can obtain the following equation of motion for this segment based on the Newton's second law in the y direction

$$F_y = F_{2y} - F_{1y} = m a_y, \quad (18)$$

where $m = \mu \Delta x$ (assuming the string can be stretched but there is no movement along the x direction), $a_y = \frac{\partial^2 y}{\partial t^2}$, and

$$F_{1y} = F_T \tan \theta_1 = F \frac{\partial y(x)}{\partial x}, \quad (19)$$

$$F_{2y} = F_T \tan \theta_2 = F \frac{\partial y(x + \Delta x)}{\partial x}, \quad (20)$$

$$F_{2y} - F_{1y} = F_T \left(\frac{\partial y(x + \Delta x)}{\partial x} - \frac{\partial y(x)}{\partial x} \right) = F_T \frac{\partial^2 y(x)}{\partial x^2} \Delta x. \quad (21)$$

We use partial derivatives since y is the function of both x and t . At the end of the day, by plugging the above results back to Eq. (18), we can obtain the wave equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{F_T} \frac{\partial^2 y}{\partial t^2}, \quad (22)$$

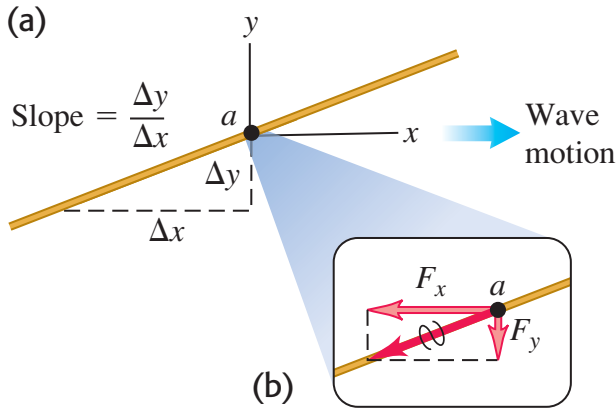
and it is straightforward to see that $y = f(x - vt)$, which describes a traveling wave with velocity v , is the solution to Eq. (22) as long as $v = \sqrt{\frac{F_T}{\mu}}$. Therefore, we can always write the wave equation as

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}. \quad (23)$$



3 Energy in Wave Motion

Traveling waves carry energy, such as the energy we received from sunlight and the destructive energy released from an earth quake. To produce any wave motion, we have to apply a force to a portion of the wave medium. As the wave propagates, each portion of the medium exerts a force and does work on the adjacent portion. In this way wave can transport energy from one region of space to another.



Let us picture a wave $y(x, t) = A \cos(kx - \omega t)$ traveling from left to right (the $+x$ direction) on a string, and consider a particular point a on the string as shown above. Let us divide the string into two parts: the part to the left of point a and the string to the right of point a . We want to compute the energy per unit time transferred from the left part to the right part. That is to say that we are focusing on the work done by the left part of the string to the right part of the string. It is also important to notice that we are considering the transverse wave, which means that every point in the string has no longitudinal motion. Therefore, the corresponding power (rate of doing work) can be written as

$$P(x, t) = F_y(x, t)v_y(x, t), \quad (24)$$

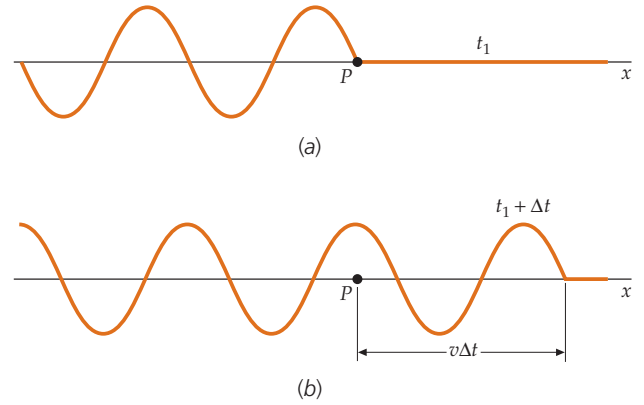
where $F_y(x, t)$ and $v_y(x, t)$ are the transverse force exerted on the right part of the string and velocity of the point a , respectively. In addition, $F_y = -\frac{\partial y}{\partial x}F_T = F_T k A \sin(kx - \omega t)$ with F_T the string tension along the x direction and $v_y(x, t) = \frac{\partial y}{\partial t} = \omega A \sin(kx - \omega t)$. As shown in the figure, we need to put a minus sign in F_y since it is negative when the slope is positive. Putting everything together, we have

$$P(x, t) = -\frac{\partial y}{\partial x}F_T \frac{\partial y}{\partial t} = F_T k \omega A^2 \sin^2(kx - \omega t) \geq 0. \quad (25)$$

Using $v = \sqrt{\frac{F_T}{\mu}}$ and $v = \frac{\lambda}{T} = \frac{\omega}{k}$, we have

$$P(x, t) = \mu v \omega^2 A^2 \sin^2(kx - \omega t), \quad (26)$$

and the average power $\langle P \rangle = \frac{1}{2} \mu v \omega^2 A^2$, because the time average of $\sin^2(kx - \omega t)$ in a period is $1/2$.



As shown above, the energy travels along a taut (tightly pulled) string at an average speed equal to the wave speed v , thus the average energy $\langle \Delta E \rangle$ flowing past point P during a time interval Δt is

$$\langle \Delta E \rangle = \langle P \rangle \Delta t = \frac{1}{2} \mu \omega^2 A^2 v \Delta t = \frac{1}{2} \mu \omega^2 A^2 \Delta x. \quad (27)$$

Note that $\mu \Delta x = \Delta m$ and therefore

$$\langle \Delta E \rangle = \frac{1}{2} \Delta m \omega^2 A^2. \quad (28)$$

Furthermore, from our last lecture, we have learnt that the average energy of a simple harmonic oscillator with mass m is $\frac{1}{2} m \omega^2 A^2$. Therefore, we can understand and interpret the effect of the energy flow in a string. The energy flowed into the right part of the string are used to provide the oscillation energy for the string, which is at rest before the energy flow arrives.

Kinetic Energy in the string

For the wave function $y(x, t) = A \cos(kx - \omega t)$, the kinetic energy dK associated with a string element of mass $dm = \mu dx$ is given by $dK = \frac{1}{2} dm v_y^2$, where v_y is the transverse speed of the oscillating string element, obtained by differentiating with respect to time while holding x constant:

$$v_y(x, t) = \frac{\partial y(x, t)}{\partial t} = \omega A \sin(kx - \omega t) \quad (29)$$

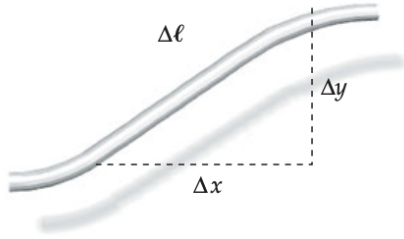
Thus, dividing dK by dt gives the rate at which kinetic energy passes through a string element, and thus the rate at which kinetic energy is carried along by the wave. The dx/dt is the wave speed v , so the rate at which kinetic energy is transported is

$$\frac{dK}{dt} = \frac{1}{2} \mu v \omega^2 A^2 \sin^2(kx - \omega t), \quad (30)$$

and its average over time is $\left(\frac{dK}{dt}\right)_{av} = \frac{1}{4} \mu v \omega^2 A^2$. It is half of the total power while the other half is stored as the potential energy.



Potential Energy in the string



Let us derive an expression for the potential energy of a segment of a string carrying a traveling wave. The potential energy of a segment equals the work done by the tension in stretching the string, which is $\Delta U = F_T(\Delta\ell - \Delta x)$, where F_T is the tension, $\Delta\ell$ is the length of the stretched segment, and Δx is its original length.

$$\begin{aligned} 1. \quad \Delta\ell &= (\Delta x^2 + \Delta y^2)^{1/2} = \Delta x(1 + (\Delta y/\Delta x)^2)^{1/2} \\ &= \Delta x \left[1 + \frac{1}{2}(\Delta y/\Delta x)^2 + \frac{1/2(1/2-1)}{1 \times 2}(\Delta y/\Delta x)^4 + \dots \right] \\ &\approx \Delta x \left[1 + \frac{1}{2}(\Delta y/\Delta x)^2 \right] \text{ in the limit of } \Delta y/\Delta x \ll 1. \\ \text{So, } \Delta\ell - \Delta x &\approx \frac{1}{2}(\Delta y/\Delta x)^2 \Delta x \Rightarrow \Delta U \approx \frac{1}{2}F_T(\Delta y/\Delta x)^2 \Delta x \end{aligned}$$

$$\begin{aligned} 2. \quad y &= A \cos(kx - \omega t) \Rightarrow \\ \partial y / \partial x &= -Ak \sin(kx - \omega t) \Rightarrow \\ \Delta U &= \frac{1}{2}F_T(\partial y / \partial x)^2 \Delta x = \frac{1}{2}F_TA^2k^2 \sin^2(kx - \omega t) \Delta x. \end{aligned}$$

Recall that the average kinetic energy and the average potential energy are equal, we expect that elastic potential energy is carried along with the wave, and at the same average rate as the kinetic energy.

Note that for a transverse wave, both potential and kinetic energies are maximum as the wave crosses the $y = 0$ point. The speed is maximum and the stretch is maximum at the $y = 0$. Both potential and kinetic energies are zero at $y = A$ point. (The speed is zero and the stretch is zero at the $y = A$ point.)

Wave Intensity

Waves on a string carry energy in just one dimension of space (along the direction of the string). But other types of waves, including sound waves in air, carry energy across all three dimensions of space. For waves that travel in three dimensions, we define the intensity (denoted as I) to be the power per unit area transported by the wave across a surface perpendicular to the direction of propagation. It is usually measured in watts per square meter (W/m^2).

If waves spread out equally in all directions from a centered source, then intensity through a sphere with radius r and surface area $4\pi r^2$ is

$$I = \frac{P}{4\pi r^2}, \quad (31)$$

which follows directly from energy conservation. This is called inverse-square law for intensity.

4 The Principle of Superposition

1. Wave superposition: The principle of superposition states that the total wave displacement at any point where two or more waves overlap is the sum of the displacement of the individual waves.

$$y(x, t) = y_1(x, t) + y_2(x, t), \quad \text{superposition} \quad (32)$$

The overlapping of waves is also called interference in physics.

2. Overlapping waves algebraically add to produce a resultant wave (or net wave).
3. Overlapping waves do not in any way alter the travel of each other.

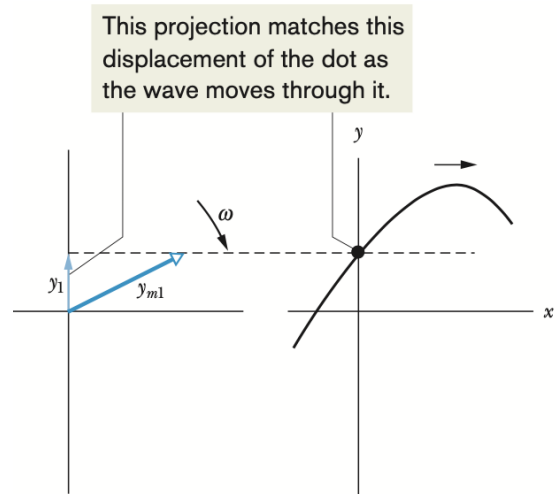
4.1 Two waves moving in the same direction and Phasor

The simplest example is to add two waves with the same amplitude and traveling in same direction with a phase shift of ϕ .

$$\begin{aligned} &A \sin(kx - \omega t) + A \sin(kx - \omega t + \phi) \\ &= \left[2A \cos \frac{\phi}{2} \right] \sin(kx - \omega t + \frac{\phi}{2}). \end{aligned} \quad (33)$$

The resultant wave is traveling in the same direction with the amplitude $2A \cos \frac{\phi}{2}$ and phase $\phi/2$. When $\phi = 0$, we have constructive interference and maximum amplitude $2A$. When $\phi = \pi$, we have destructive interference and minimum amplitude 0. For more complicated case, we need to use phasor (phase vector).

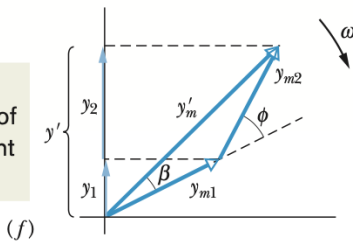
A phasor is a vector that rotates around its tail, which is pivoted at the origin of a coordinate system. The magnitude of the vector is equal to the amplitude of the wave that it represents. The angular speed of the rotation is equal to the angular frequency ω of the wave. For example, the wave $y = y_{m1} \sin(kx - \omega t)$ is represented by the phasor shown below. A second wave that **lags behind by ϕ (delayed in time)** can be written as $y = y_{m2} \sin(kx - \omega t + \phi)$.





Adding the two phasors as vectors gives the resultant phasor of the resultant wave.

This is the projection of the resultant phasor.



When two waves travel along the same string in the same direction, we can represent them and their resultant wave in a phasor diagram. We can use phasors to combine waves even if their **amplitudes** are different.

To understand this method further, one may adopt the complex description of waves as follows. Let the wave function be

$$\tilde{y}(x, t) = Ae^{i\theta} = A \cos \theta + iA \sin \theta, \quad \text{with } \theta = kx - \omega t. \quad (34)$$

The real part of $\tilde{y}(x, t)$ is a cosine wave while the imaginary part gives a sine wave. It is important to note that the real and imaginary parts are independent when two waves are added together.

Suppose we consider two waves (moving with the same speed and frequency) summing together as follows

$$A_1 e^{i\phi_1} + A_2 e^{i\phi_2} = A e^{i\beta + i\phi_1}, \quad \text{with } \phi_2 - \phi_1 = \phi = \text{Const.} \quad (35)$$

By comparing real and imaginary parts of the above equation, one finds

$$A \cos \beta = A_1 + A_2 \cos \phi \quad (36)$$

$$A \sin \beta = A_2 \sin \phi. \quad (37)$$

It is then straightforward to solve for the amplitude and phase of the resultant sum of two waves traveling in the same direction.

4.2 Two waves moving in opposite direction and Standing Wave

Standing wave on a string: For a given string, there are certain frequencies for which superposition results in a stationary vibration pattern called a standing wave. (Standing wave is no longer traveling. To emphasize the

difference, a wave that does move along the string is called a traveling wave.) Let us consider the superposition of two waves

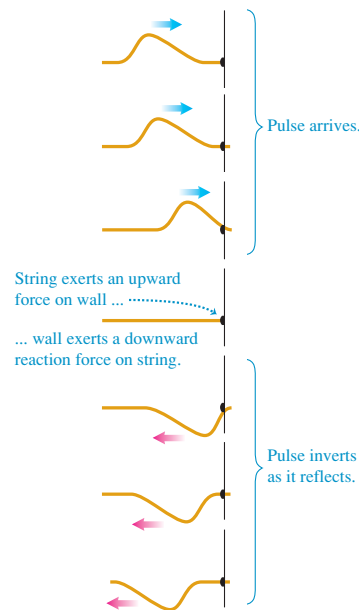
$$y_1(x, t) = A \cos(kx - \omega t), \quad \text{right-moving (+x) direction,}$$

$$y_2(x, t) = -A \cos(kx + \omega t), \quad \text{left-moving (-x) direction.}$$

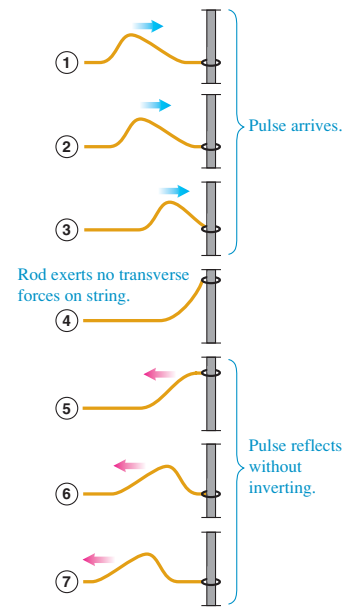
In general, $y_2(x, t)$ could be the reflected wave when $y_1(x, t)$ reaches a fixed boundary of the medium. Let us first try to understand the effect of boundary below.

4.2.1 understanding wave reflection at boundaries

(a) Wave reflects from a fixed end.



(b) Wave reflects from a free end.



In fact, $y_2(x, t)$ can be viewed as the reflected wave after the incident right wave $y_1(x, t)$ reflects from a fixed end. Note also that the change in sign corresponds to a shift in phase of 180° or π radians due to the reaction force from the fixed end during the reflection. When the wave arrives at the fixed end, it exerts a force to the fixed end, and the reaction to this force from the fixed end (is opposite to the action based on Newton's third law), "kicks back" on the string and sets up a reflected wave in the reverse direction. In contrast, wave reflected from a free end has no such phase shift.²

²Mathematical interpretation:

- In math, one can look at this in terms of **Dirichlet boundary condition** ($y = 0$ at $x = 0$ for any t) by constructing an odd function about the reflection point. Let us suppose the reflection occurs at the $x = 0$ point. As a result of the superposition of the incident (incoming) wave and reflected (outgoing) wave, the displacement y has to be zero at all time. Thus the total displacement must be

$$y(x, t) = f(x - vt) - f(-x - vt), \quad \text{which is odd under the exchange } x \rightarrow -x \text{ and remains zero at any time.} \quad (38)$$

Here we suppose $f(x - vt)$ is the wavefunction of the incoming wave, thus the outgoing wavefunction is then $-f(-x - vt)$, where the argument $-x - vt$ indicates the opposite propagation. The minus sign (-) between incoming and outgoing wave before and after the reflection is necessary to ensure the fixed hard boundary.

- In contrast, there is a free end boundary condition called **Neumann boundary condition** ($\frac{\partial y}{\partial x} = 0$ at $x = 0$ for any t). Since $F_y = F_x \frac{\partial y}{\partial x}$, one finds there is no vertical force applied at the boundary. In physics, this can be realized by imposing a free end (loop with zero mass). Mathematically, this leads to the symmetric even function about $x = 0$, namely,

$$y(x, t) = f(x - vt) + f(-x - vt), \quad \text{which is even under the exchange } x \rightarrow -x \text{ and } \frac{\partial y}{\partial x} \text{ remains zero at any time.} \quad (39)$$



4.2.2 Standing Wave

The resulting superposition of these two waves is then

$$\begin{aligned} y(x, t) &= A[\cos(kx - \omega t) - \cos(kx + \omega t)] \\ &= (2A \sin kx) \sin \omega t = (A_{SW} \sin kx) \sin \omega t, \end{aligned} \quad (40)$$

where the amplitude of the standing wave $A_{SW} = 2A$. The above equation has two factors: a function of x and a function of t . The factor $A_{SW} \sin kx$ shows that at each moment the shape of the string is a sine curve. But unlike a wave traveling along a string, the wave shape stays in the same position, oscillating up and down as described the $\sin \omega t$ factor.

We can find that the amplitude of the oscillation $A_{SW} \sin kx$ is always zero as long as $kx = n\pi$ with n an integer. These places are the nodes. At a node the displacements of the two waves are always equal and opposite and cancel each other out, which is known as destructive interference. Midway between two adjacent nodes are the points of greatest amplitude given by $kx = n\pi + \frac{\pi}{2}$. They are called antinodes due to constructive interferences of two waves. The distance between successive nodes or between successive antinodes is one half-wavelength, or $\lambda/2$.

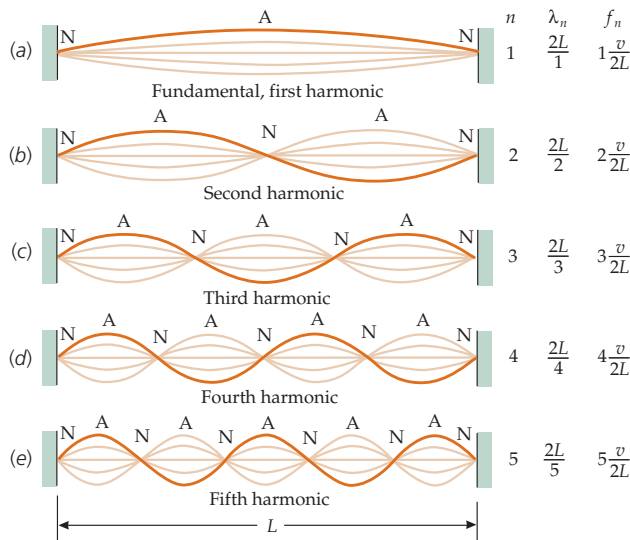


FIGURE 16-12 Standing waves on a string that is fixed at both ends. Antinodes are labeled A and nodes are labeled N. The n th harmonic has n antinodes, where $n = 1, 2, 3, \dots$

Let us consider a string of a definite length L , rigidly fixed at both ends. Such strings are found in many musical instruments. Only standing waves with certain wavelengths can form on this string. We can also see from above figure that the standing wave must have a node at both end of the string due to fixed boundary condition. Therefore we have the following standing wave condition

$$L = n \frac{\lambda_n}{2}, \quad n = 1, 2, 3, \dots \quad (41)$$

The corresponding frequency then can be computed by

$$f_n = \frac{1}{T_n} = \frac{v}{\lambda_n} = n \frac{v}{2L}, \quad n = 1, 2, 3, \dots, \quad (42)$$

where $v = \sqrt{\frac{F_T}{\mu}}$ is the velocity of the wave. Each frequency with its associated vibration pattern is called a normal mode. The lowest frequency f_1 is called the fundamental frequency.

As opposed to the traveling wave, **standing waves do NOT transfer energy** from one end to another, since equal amount of energy flows carried by the incident wave and reflected wave cancel each other. (The mechanical energy flows back and forth from a node to an antinode as the string goes from maximum curvature to straight, and does not propagate along the string.)

5 Sound Wave and Longitudinal Wave

Sound waves (with the audible frequency range of f from 20 to 20,000 Hz) in the air and liquid (fluid in general) are in the form of longitudinal waves, since the fluid does not normally support shear. In contrast, sound can travel in solids in terms of both transverse and longitudinal waves.

In gases, the sound wave propagates by compressing and expanding the gas and changing the pressure accordingly.

First, consider the sound wave in a pipe and let the displacement of the element of a fluid along the pipe (the direction of wave propagation) be

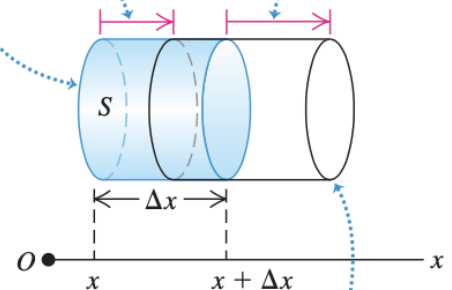
$$y(x, t) = A \cos(kx - \omega t). \quad (43)$$

5.1 Sound Waves As Pressure Fluctuations

Consider a volume of gas with the volume $V = S\Delta x$.

Undisturbed cylinder of fluid has cross-sectional area S , length Δx , and volume $S\Delta x$.

A sound wave displaces the left end of the cylinder by $y_1 = y(x, t)$... and the right end by $y_2 = y(x + \Delta x, t)$.



The change in volume of the disturbed cylinder of fluid is $S(y_2 - y_1)$.

As a sound wave propagates along the x -axis, the left and right ends undergo different displacements y_1 and y_2 . The change of the volume then reads

$$\Delta V = S[y(x + \Delta x, t) - y(x, t)] = S \Delta y|_{\text{fixed } t}. \quad (44)$$



Recall the Bulk modulus, the elastic property of liquid under pressure, which reads

$$B = -\frac{\Delta P}{\Delta V/V}. \quad (45)$$

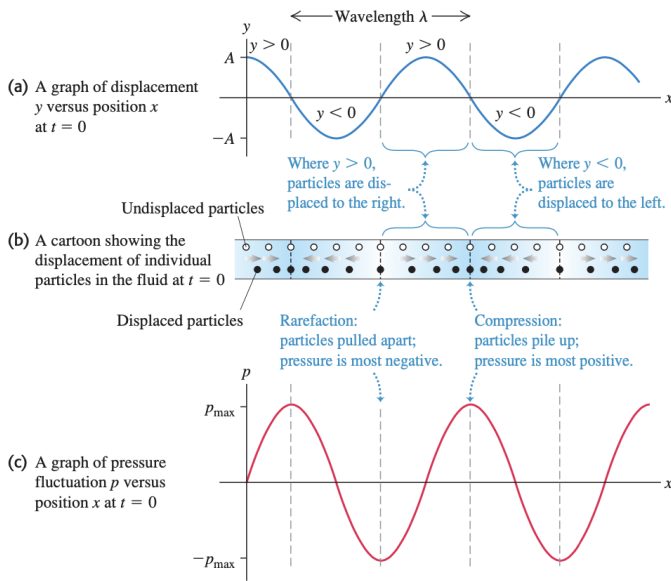
When sounds travel through medium, they create pressure difference (on top of the equilibrium pressure P_0), ΔP . Let us solve for the pressure fluctuation ΔP , we have

$$\Delta P = -B \frac{\Delta V}{V} = -B \frac{\Delta y}{\Delta x} \Big|_{\text{fixed } t} = -B \frac{\partial y}{\partial x}, \quad (46)$$

and after the differentiation on $y(x, t) = A \cos(kx - \omega t)$

$$\Delta P = \Delta P_m \sin(kx - \omega t), \quad (47)$$

with the amplitude of the **pressure fluctuation (variation)** $\Delta P_m = BAK$.



The comparison between y and ΔP is shown above for illustration purpose.

Deriving the wave speed

- It is then straightforward to consider the motion of the gas with the volume $V = S\Delta x$ based on Newton's 2nd law and derive the wave equation as follows. Suppose $\Delta P_2 = -B \frac{\partial y}{\partial x}$ at $x + \Delta x$ and $\Delta P_1 = -B \frac{\partial y}{\partial x}$ at x , then the total pressure force on the gas is given by

$$A(\Delta P_1 - \Delta P_2) = BA \frac{\partial^2 y}{\partial x^2} \Delta x. \quad (48)$$

According to Newton's 2nd law, it provides the acceleration for the mass $\Delta m = \rho A \Delta x$

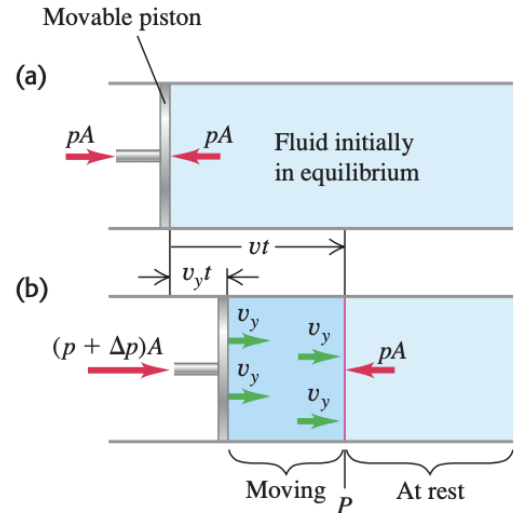
$$BA \frac{\partial^2 y}{\partial x^2} \Delta x = \Delta m a = \Delta m \frac{\partial^2 y}{\partial t^2}, \quad (49)$$

which yields the wave equation for sound waves

$$\frac{\partial^2 y}{\partial t^2} = \frac{B}{\rho} \frac{\partial^2 y}{\partial x^2}, \quad (50)$$

with the wave propagation speed $v = \sqrt{\frac{B}{\rho}}$. As previously mentioned in transverse wave, the wave speed is determined by the restoring elastic property (B) and inertia ρ together.

- Second method uses the wave front and more physical consideration. A sound wave propagating in a fluid confined to a tube. (a) Fluid in equilibrium. (b) A time t after the piston begins moving to the right at speed v_y , the fluid between the piston and point P is in motion. The speed of sound waves is v .



Thus, the compression ratio is then $-(v_y A t)/(v A t) = -v_y/v$. It leads to

$$\Delta P = B \frac{v_y}{v}. \quad (51)$$

Similarly, one uses the impulse-momentum theorem and write

$$\Delta P A t = (\Delta m) v_y = \rho v t A v_y, \quad (52)$$

which gives $v^2 = B/\rho$, same as above.

- Speed of sound in the air. In HW8, we have discussed the bulk modulus for the air. Newton supposed that the air is isothermal (constant temperature) when sound propagates, and found $B = P$ with P the pressure of the air. The resulting speed is 300 m/s at room temperature. Later, Laplace realized that the process should be adiabatic (no heat transfer since the compression and propagation happen too fast), and then obtained $B = \gamma P$ with $\gamma = 7/5$. One gets $v = 340$ m/s (sometimes people use 344 m/s) for the **speed of sound** and a better agreement with the measurement.

5.2 Interference

Consider the superposition (sum) of two sound waves from two sources S_1 and S_2 with the path lengths L_1 and L_2 , respectively.

$$y_1 = A \cos(kL_1 - \omega t), \quad (53)$$

$$y_2 = A \cos(kL_2 - \omega t). \quad (54)$$

The resultant total displacement reads

$$y = 2A \cos \frac{k\Delta L}{2} \cos \left(k \frac{L_1 + L_2}{2} - \omega t \right) \quad (55)$$

with the path length difference $\Delta L = |L_2 - L_1|$ and the total amplitude $\left| 2A \cos \frac{k\Delta L}{2} \right|$. It gives



- **Constructive interference** when

$$\frac{k\Delta L}{2} = n\pi \quad \text{or} \quad \Delta L = n\lambda;$$

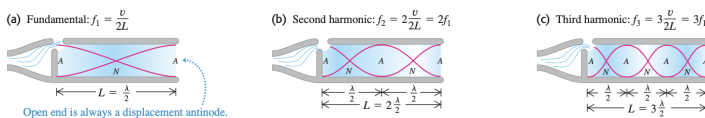
- **Destructive interference** when

$$\frac{k\Delta L}{2} = n\pi + \frac{\pi}{2} \quad \text{or} \quad \Delta L = \frac{2n+1}{2}\lambda.$$

5.3 Standing Sound Waves: A tale of Two Pipes (Open vs Closed)

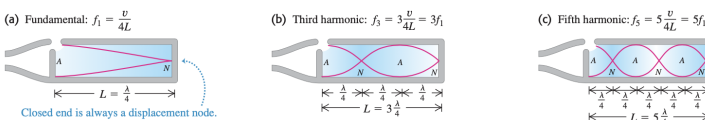
When it comes to standing sound waves, there are two different boundary conditions (B.C.s), namely, the open (free end, Neumann) and closed (fixed end, Dirichlet) B.C.s. It is important to note that these two B.C.s correspond to the antinode and node in displacement of air particles, respectively. The most important application of standing sound waves is the musical instruments.

- The **open end** is always an **antinode** in displacement, indicated by the maximum amplitude with $\frac{\partial y}{\partial x} = 0$ (This is described by **Neumann B.C.**). Since $\Delta P = -B \frac{\partial y}{\partial x}$, this implies the open end is then a node in pressure. This is expected, as open ends directly connect to the open atmosphere with a constant pressure P_0 .
- The **closed end** (stopped pipe) is always a **node in displacement** ($y = 0$ at all times described by **Dirichlet B.C.**), corresponding to an antinode in pressure.
- A **pressure node (antinode)** is always a **displacement antinode (node)**.



A **normal mode** of an oscillating system is a motion in which all particles of the system move with the **same frequency**. In the open pipe, the normal modes are characterized by the frequencies

$$f_n = \frac{nv}{2L} = nf_1, \quad \text{with } n = 1, 2, 3, \dots \quad (56)$$



Only odd harmonics are possible in a stopped pipe at one end. The above figure is showing the first three normal modes as well as the displacement nodes and antinodes.

$$f_n = \frac{nv}{4L}, \quad \text{with } n = 1, 3, 5, \dots \quad (57)$$

Resonance: Imagine a loudspeaker is used to send sound waves into a pipe. If the frequency is precisely

equal to a normal-mode frequency, the system is in resonance, and the amplitude of the forced oscillation is maximum. If there were no friction or other energy-dissipating mechanism, a driving force at a normal-mode frequency would continue to add energy to the system, and the amplitude would increase indefinitely. In such an idealized case, the amplitude of the sound wave would be infinitely high. But in any real system there is always some dissipation of energy, or damping, the amplitude of oscillation in resonance may be large, but it can not be infinite.

5.4 Sound Intensity and Sound Levels

As we mentioned earlier, it is important to note that the power transmitted in the wave is proportional to the amplitude square A^2 of the wave. This is true for all waves.

Let us consider a 3D sound wave emitted uniformly in all directions from the source. Recall that the intensity $I = P/(4\pi r^2)$ is the power per unit area.

For sound wave, the pressure change is radial, thus

$$I = \frac{Fv}{4\pi r^2} = \frac{4\pi r^2 \Delta P v_y}{4\pi r^2} = \Delta P v_y = A^2 B k \omega \sin^2(kr - \omega t). \quad (58)$$

with $v_y = A\omega \sin(kr - \omega t)$ as the radial oscillation velocity. v_y is derived similarly from $\partial y/\partial t$ from longitudinal (radial) displacement.

Then the average intensity over time is given by

$$\langle I \rangle = \frac{1}{2} A^2 B k \omega = \frac{\Delta P_m^2}{2} \frac{\omega}{Bk} = \frac{\Delta P_m^2}{2} \frac{v}{B} = \frac{\Delta P_m^2}{2} \frac{1}{\rho v}, \quad (59)$$

where $v = \sqrt{B/\rho}$ is used and $\Delta P_m = ABk$ is the amplitude of the pressure variation.

Sound level and loudness: Human ears are sensitive to the intensity of the sound wave (due to pressure variations) over a broad range. Thus, a logarithmic intensity scale is usually used. The sound intensity level β of a sound wave is defined by the equation

$$\beta = (10 \text{ dB}) \log_{10} \frac{I}{I_0}, \quad (60)$$

where I_0 is a reference intensity, chosen to be 10^{-12} W/m^2 , approximately the threshold of human hearing at 1000 Hz. Sound intensity levels are expressed in decibels, abbreviated **dB**. At the threshold, the sound intensity level is **0dB**. The following table gives the sound intensity levels in decibels of some familiar sounds.

Source or Description of Sound	Sound Intensity Level, β (dB)	Intensity, I (W/m^2)
Military jet aircraft 30 m away	140	10^2
Threshold of pain	120	1
Riveter	95	3.2×10^{-3}
Elevated train	90	10^{-3}
Busy street traffic	70	10^{-5}
Ordinary conversation	65	3.2×10^{-6}
Quiet automobile	50	10^{-7}
Quiet radio in home	40	10^{-8}
Average whisper	20	10^{-10}
Rustle of leaves	10	10^{-11}
Threshold of hearing at 1000 Hz	0	10^{-12}



5.5 Beat

Consider the superposition of two waves with similar frequencies

$$y_1 = A \cos(kx - \omega_1 t), \quad (61)$$

$$y_2 = A \cos(kx - \omega_2 t) \quad (62)$$

with $\bar{\omega} \equiv \frac{\omega_1 + \omega_2}{2}$, $\Delta\omega \equiv \omega_1 - \omega_2$, and $\bar{\omega} \gg |\Delta\omega|$. It is then easy to find

$$y(x, t) = y_1 + y_2 \quad (63)$$

$$= 2A \cos\left(\frac{\Delta\omega t}{2}\right) \cos(kx - \bar{\omega}t). \quad (64)$$

Computer the period of the sound intensity which is given by the square of the amplitude

$$\left[2A \cos\left(\frac{\Delta\omega t}{2}\right)\right]^2 = 2A^2(1 + \cos \Delta\omega t),$$

which has a period of $T = 2\pi/\Delta\omega$. Thus one can normally define the so-called **beat frequency**

$$\Delta f = \frac{\Delta\omega}{2\pi} = \left|\frac{\omega_1}{2\pi} - \frac{\omega_2}{2\pi}\right| = |f_1 - f_2|. \quad (65)$$



If the two propellers on this airplane are not precisely synchronized, the pilots, passengers, and listeners on the ground will hear beats.

5.6 Doppler Effect

Doppler Effect: Choosing medium (air or liquid water) as the reference frame, there are **motion related frequency changes** given by

$$f' = f \frac{v \pm v_D}{v \pm v_S}, \quad (66)$$

where

- v is the **speed of sound in the medium** (air, 344 m/s);
- v_D is the detector's speed **relative to the air**;
- v_S is the source's **speed relative to the air**.

The choice of plus or minus sign is set by a simple rule: f' **increases (decreases)** if the detector and source are **moving closer (further away)** to each other.

Explanation of the Doppler effect:

- Normally, when both Source and Detector are at rest, we see that the essence of the frequency is simply giving the **number of cycles** (full waves of ups and downs in amplitude) per unit time

$$f = \frac{1}{T} = \frac{v}{\lambda} = \frac{v \Delta t}{\lambda \Delta t}. \quad (67)$$

Here the time interval Δt is arbitrary, and can be just set to 1 second.

- If the detector (listener, receiver) moves towards the source, $v \Delta t$ in the above equation is then changed accordingly to $(v + v_D) \Delta t$. Therefore,

$$f' = \frac{(v + v_D) \Delta t}{\lambda} \frac{1}{\Delta t} = \frac{v + v_D}{v} f. \quad (68)$$

- If the source moves with a speed v_S towards the detector, then the wavelength of the sound wave gets changed to $\lambda' = \lambda - v_S T = (v - v_S) T$. Hence one finds

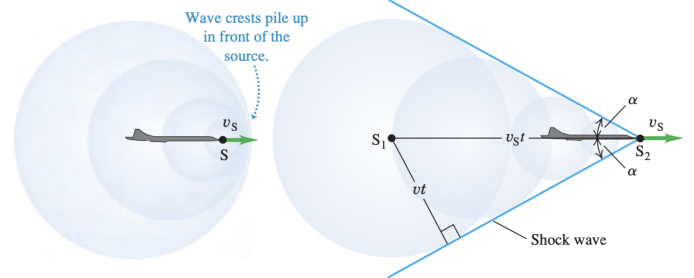
$$f' = \frac{v}{\lambda'} = \frac{v}{v - v_S} \frac{1}{T} = \frac{v}{v - v_S} f. \quad (69)$$

- In the above two cases, frequencies detected are higher than the original f since D and S are moving closer to each other. If they are moving away from each other, one only needs to flip the sign in the above equations and obtains a reduced frequency instead.

5.7 Supersonic Waves and Shock Waves

(a) Sound source S (airplane) moving at nearly the speed of sound

(b) Sound source moving faster than the speed of sound



Let us consider the 3D wave fronts created by a moving sound source with speed v_S . If $v_S < v$, one sees that new fronts are always inside old ones as shown in the figure (a) above, although the wave fronts start to pile up in front of the moving source and create extra drag force.

When $v_S > v$, which means source is traveling at a supersonic speed, Doppler effect no longer describe the sound wave in front of the source. Since the wave fronts are traveling at speed v , the new fronts are now always **created (pulled out) ahead** of the old fronts along the direction of the motion. Now the shape of the wave front is changed from sphere to cone, which is also called shock wave. Using simple geometry, one can find the **Mach cone** angle α

$$\sin \alpha = \frac{vt}{v_S t} = \frac{v}{v_S} < 1. \quad (70)$$

The ratio $v_S > v$ is called the **Mach number**, which is greater than 1 for all supersonic speeds, and $\sin \alpha$ is the reciprocal of the Mach number.

Supersonic jets often creates super loud sonic booms and shock waves due to the quick rise (by pushing the air in front of the jet) and fall (by leaving a vacuum behind) of air pressure.